# HP PREDNISOLONE TABLET 5 mg

VIPRExx-02

## DESCRIPTION
Round, white uncoated tablet, shallow convex faces; "HD" embossed and scored on the same face.

## COMPOSITION
Prednisolone 5 mg/tablet.

## ACTIONS AND PHARMACOLOGY
Prednisolone, a synthetic glucocorticoid, has both anti-inflammatory and immunosuppressant actions. It acts by inhibition of phagocytosis, leukocyte migration and capillary dilatation, and prevention or suppression of cell mediated (delayed hypersensitivity) immune reactions respectively. It is readily absorbed orally, metabolised in the liver and excreted via the kidneys.

## INDICATIONS
Prednisolone is used in the treatment of conditions where a routine systemic corticosteroid therapy is indicated. Its weaker sodium-retaining action usually makes it more suitable than cortisone in such conditions as rheumatoid arthritis, rheumatic fever, status asthmaticus and ulcerative colitis.

## CONTRAINDICATION
Use is not recommended in nursing mothers, wherever possible.

## WARNING AND PRECAUTIONS
- Caution in patients with osteoporosis, peptic ulcer, psychoses, systemic fungal infections, diabetes mellitus, hypertension, myasthenia gravis, ocular herpes simplex, glaucoma, hypothyroidism, history of tuberculosis, renal and hepatic function impairment.
- When medication is to be discontinued, dosage should be reduced gradually. Abrupt cessation of prolonged therapy may produce acute adrenal insufficiency.
- Frequent monitoring of drug effect is required.
- Caution in receiving vaccinations, other immunizations and skin test.
- Children on prolonged therapy should be closely observed.
- Safety for use in pregnancy has not been established.
- Pheochromocytoma crisis, which may be fatal, has been reported after administration of systemic corticosteroids. Corticosteroids should only be administered to patients with suspected or identified pheochromocytoma after an appropriate risk / benefit evaluation.
- Scleroderma renal crisis: Caution is required in patients with systemic sclerosis because of an increased incidence of (possibly fatal) scleroderma renal crisis with hypertension and decreased urinary output observed with a daily dose of 15 mg or more prednisolone.

## PREGNANCY AND LACTATION
**Pregnancy:** The ability of corticosteroids to cross the placenta varies between individual drugs, however, 88 % of prednisolone is inactivated as it crosses the placenta.

There is no evidence that corticosteroids result in an increased incidence of congenital abnormalities, such as cleft palate/lip in man. However, when administered for prolonged periods or repeatedly during pregnancy, corticosteroids may increase the risk of intra-uterine growth retardation. The use of corticosteroids, including prednisolone, during pregnancy may also result in stillbirth. Hypoadrenalism may, in theory, occur in the neonate following prenatal exposure to corticosteroids but usually resolves spontaneously following birth and is rarely clinically important. As with all drugs, corticosteroids should only be prescribed when the benefits to the mother and child outweigh the risks. When corticosteroids are essential however, patients with normal pregnancies may be treated as though they were in the non-gravid state. Patients with pre-eclampsia or fluid retention require dose monitoring.

**Lactation:** Corticosteroids are excreted in small amounts in breast milk. However, doses of up to 40 mg daily of prednisolone are unlikely to cause systemic effects in the infant. Infants of mothers receiving 40 mg or more daily should be monitored for signs of adrenal suppression but the benefits of breast- feeding are likely to outweigh any theoretical risk.

## MAIN SIDE/ADVERSE EFFECTS
**Endocrine/metabolic:** Suppression of the hypothalamic-pituitary-adrenal axis, growth suppression in infancy, childhood and adolescence, menstrual irregularity and amenorrhoea. Hirsutism, weight gain, impaired carbohydrate tolerance, hyperglycaemia, precipitation of diabetes mellitus or increased requirement for anti-diabetic therapy in pre-existing diabetes. Negative nitrogen and calcium balance. Increased appetite. Hypercholesterolaemia and hypertriglyceridaemia. High doses or prolonged administration of corticosteroids may cause Cushing's syndrome.

**Anti-inflammatory and immunosuppressive effects:** Increased susceptibility and severity of infections with suppression of clinical symptoms and signs, opportunistic infections, recurrence of dormant tuberculosis, candidiasis.

**Musculoskeletal:** Osteoporosis, vertebral and long bone fractures, tendon rupture. Proximal myopathy, muscular weakness, muscular atrophy and buffalo hump. Avascular osteonecrosis has been associated with long term or high dose corticosteroid therapy.

**Fluid and electrolyte disturbance:** Sodium and water retention, potassium loss, hypokalaemic alkalosis, oedema.

**Blood and lymphatic system disorders:** Corticosteroids have the potential to increase the coagulability of blood.

**Vascular disorders:** Hypertension.

**Neuropsychiatric:** A wide range of psychiatric reactions including affective disorders (such as irritable, euphoric, depressed and labile mood, and suicidal thoughts), psychotic reactions (including mania, delusions, hallucinations, and aggravation of schizophrenia), behavioural disturbances, irritability, anxiety, sleep disturbances, and cognitive dysfunction including confusion and amnesia have been reported. Reactions are common and may occur in both adults and children. In adults, the frequency of severe reactions has been estimated to be 5 % - 6 %. Psychological effects have been reported on withdrawal of corticosteroids; the frequency is unknown. Psychological dependence, depression, insomnia, psychosis, delirium, and nervousness and/or restlessness have also been reported.

Increased intra-cranial pressure with papilloedema in children (pseudotumour cerebri), usually after treatment withdrawal. Aggravation of epilepsy.

**Ophthalmic:** Increased intra-ocular pressure, glaucoma, papilloedema, posterior subcapsular cataracts, corneal or scleral thinning, serious retinal detachment, exacerbation of ophthalmic viral of fungal diseases and sudden blindness. Potential for exophthalmos to occur with long-term administration of corticosteroids.

**Gastrointestinal:** dyspepsia, peptic ulceration with perforation and haemorrhage, acute pancreatitis, candidiasis, hiccups and nausea.

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**Dermatological:** Impaired healing, skin atrophy, bruising, telangiectasia, striae, acne, skin thinning, flushing, hyperhidrosis.

**General:** Hypersensitivity including anaphylaxis, has been reported. Leucocytosis. Thromboembolism, myocardial rupture following recent myocardial infarction, increased risk of Stevens-Johnson syndrome or toxic epidermal necrolysis, tumour lysis syndrome.

**Withdrawal symptoms and signs:** Too rapid a reduction of corticosteroid dosage following prolonged treatment can lead to acute adrenal insufficiency, hypotension and death. A 'withdrawal syndrome' may also occur including, fever, myalgia, weakness, arthralgia, rhinitis, conjunctivitis, painful itchy skin nodules, loss of weight, mental changes, emotional changes, nausea, vomiting, hypotension, benign intracranial hypertension, dizziness, headache, and reappearance of disease symptoms.

## DRUG INTERACTIONS

- **Hepatic microsomal enzyme inducers:** Rifampicin, rifabutin, carbamazepine, phenobarbitone, phenytoin, primidone and aminoglutethimide enhance the metabolism of prednisolone and its therapeutic effects may be reduced.
- **Anticoagulants:** The efficacy of coumarin anticoagulants may be enhanced or reduced by concomitant corticosteroid therapy and close monitoring of the INR or prothrombin time is required to avoid spontaneous bleeding.
- **Non-steroidal anti-inflammatory drugs (NSAIDs):** There is an increased risk of gastro-intestinal bleeding and ulceration when corticosteroids are used concomitantly with NSAIDs, including aspirin. The renal clearance of salicylates is increased by corticosteroids and steroid withdrawal may result in salicylate intoxication.
- **Hormones:** Oestrogens may enhance the effects of corticosteroids and dosage adjustments may be required in some cases.
- Prednisolone can inhibit the growth stimulating effect of somatropin.
- **Antibacterials:** Corticosteroids can lower plasma concentrations of isoniazid and enhance its renal clearance.
- **Antifungals:** Concomitant use with ketoconazole may inhibit the metabolism of prednisolone and enhance its adrenal suppressive effects. There is the potential for an increased risk of hypokalaemia when corticosteroids are used concomitantly with amphotericin.
- **Antineoplastics and immunosuppressants:** Mutual inhibition of metabolism may occur between ciclosporin and prednisolone, and may increase the plasma concentration of either drug.
- **Antivirals:** Plasma concentrations of prednisolone may be increased with antiviral drugs such as ritonavir and indinavir.
- **Vaccines:** Concomitant use of high dose corticosteroids and live vaccines should be avoided. Corticosteroids may impair the immune response to other vaccines.
- **Neuromuscular blockers:** Neuromuscular blocking effects may be antagonised by prednisolone in patients with adrenocortical insufficiency.
- **Cardiac glycosides:** There is the potential for an increased risk of digitalis toxicity associated with hypokalaemia, when corticosteroids and cardiac glycosides are used concomitantly.
- **Sympathomimetics:** There is an increased risk of hypokalaemia if high doses of corticosteroids are given with high doses of bambuterol, fenoterol, formoterol, reproterol, ritodrine, salbutamol, salmeterol, terbutaline and tulobuterol.
- **Thalidomide:** The dosage of prednisolone needs to be reduced considerably when used with thalidomide. It has been suggested that prednisolone should not be given with thalidomide.
- **Intra-uterine devices (IUDs):** There is the potential for contraceptive failure in women using intra-uterine devices and receiving corticosteroid therapy.
- **Other:** The desired effects of hypoglycaemic agents (including insulin), antihypertensives and diuretics are antagonised by corticosteroids and the hypokalaemic effects of acetazolamide loop diuretics, and thiazide diuretics are enhanced. The effect of corticosteroids may be reduced for 3 to 4 days after the use of mifepristone. There is the potential for an increased risk of hypokalaemia when corticosteroids and theophylline are used concomitantly.

## OVERDOSE

Clinical features: Nausea and vomiting, hyperglycaemia, occasional gastrointestinal bleeding.

Treat overdosage by symptomatic measures.

## DOSAGE AND ADMINISTRATION

**Adults :** Oral, 5 mg to 60 mg a day as a single dose or in divided doses, not exceeding 250 mg daily.

**Children :** As directed by physician.

The paediatric dosage is determined more by the severity of the condition and response of the patient than by age or body weight.

Note: The information given here is limited. For further information consult your doctor orpharmacist.

Storage: Store below 30˚C. Protect from light.

Presentation/Packing: Blister of 10 x 10's

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